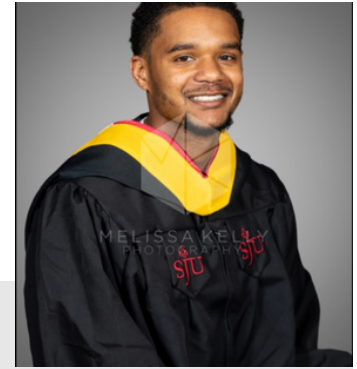


DARIOUS BROWN

AI & Data Scientist | Biotech Manufacturing & Healthcare Analytics

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Portfolio: <https://dare215.github.io/DariusBrown-Portfolio/> | www.linkedin.com/in/dariusbrown

PROFILE

AI & Data Scientist with doctoral-level training in Artificial Intelligence, Machine Learning, and Business Administration, combining advanced analytical capabilities with hands-on experience in GMP-regulated biopharmaceutical manufacturing environments. Proven ability to design, develop, and implement predictive models that enhance quality systems, reduce operational risk, and improve compliance outcomes. Specializes in healthcare analytics, process optimization, and translating manufacturing data into actionable insights within ISO-5/ISO-7 cleanroom environments.

EDUCATION

Walsh College
(Expected APR28)

PhD in Artificial Intelligence & ML Specialization

Walsh College
(Expected APR28)

Doctorate of Business Administration
• Focused in Artificial Intelligence

Bellevue University
2025

Masters in Data Science

Saint Joseph's
University
2023

Bachelors in Biomedical Sciences

WORK EXPERIENCE

Invance
2024-Present

Aseptic Technician 2

- Execute GMP-compliant autologous TIL cell therapy manufacturing within ISO-5/ISO-7 cleanroom environments, ensuring end-to-end traceability, sterility assurance, and regulatory compliance across high-risk production workflows
- Perform environmental monitoring (viable and non-viable sampling) and conduct trend analysis to detect contamination patterns, supporting proactive risk mitigation and continuous quality improvement initiatives
- Designed and implemented structured data tracking frameworks to capture process variability, enabling data-driven visibility into manufacturing performance and readiness for predictive analytics integration
- Translate manufacturing and environmental datasets into actionable insights that support deviation reduction, process consistency, and operational decision-making
- Collaborate cross-functionally with Quality, MSAT, and Supply Chain teams to ensure data integrity, regulatory adherence, and alignment of process improvements with GMP standards
- Contribute to development of an AI-driven waste tracking system (dissertation-aligned) incorporating anomaly detection and forecasting within GMP and compliance constraints
- Support continuous improvement initiatives through operational trend analysis, process variability tracking, and identification of inefficiencies in material usage and workflow execution, aligning operations with quality and cost-optimization goals
- Applied structured operational analytics to identify workflow inefficiencies, support process optimization efforts, and improve manufacturing visibility within GMP-regulated environments.

2023-2024

- Executed GMP-aligned molecular biology workflows (PCR, DNA cloning, electrophoresis) within regulated laboratory environments, ensuring reproducibility and compliance
- Improved operational efficiency by ~15% through workflow optimization and process standardization, reducing variability and increasing throughput
- Captured, cleaned, and structured experimental datasets to enhance traceability, reproducibility, and downstream analytical readiness
- Maintained strict adherence to SOPs and regulatory requirements, ensuring data integrity, contamination control, and audit readiness
- Supported data-driven laboratory operations by aligning experimental execution with quality metrics and reporting standards

TECHNICAL SKILLS

- Languages: Python, R, SQL
- Frameworks: Scikit-learn, PyTorch
- Tools: Streamlit, MLflow, Git, Jupyter Notebook
- ML Techniques: Random Forest, Logistic Regression, LSTM, CNNs, Clustering
- Analytics: Time-Series Forecasting, Statistical Modeling, Hypothesis Testing, Predictive Analytics, Feature Engineering
- Explainability: SHAP, Feature Importance, Model Diagnostics
- Domain: Biotech Manufacturing (GMP), Healthcare Analytics, Financial Risk Modeling

AI PROJECTS & SYSTEMS

- Loan Default Prediction System (Deployed)
 - Developed supervised machine learning models (Random Forest, Logistic Regression) for credit risk prediction
 - Achieved strong predictive performance using ROC-AUC and classification metrics
 - Integrated SHAP explainability to enhance model transparency
 - Deployed Streamlit-based application supporting real-time and batch inference
- Material Waste Tracker (Dissertation Project)
 - Designed AI-ready data architecture for GMP-regulated environments
 - Built Streamlit application with anomaly detection, forecasting, and reporting capabilities
 - Developed predictive framework targeting 10-25% waste reduction
 - Implemented audit-compliant data logging and de-identification protocols
- Bitcoin Price Forecasting (LSTM)
 - Built LSTM deep learning model for time-series forecasting
 - Conducted EDA, PMF/CDF analysis, and regression modeling
 - Identified key drivers including trading volume and market capitalization
- Generative AI Hallucination Detection
 - Fine-tuned GPT-based models to detect hallucinations using confidence scoring
 - Improved reliability and evaluation of generative AI systems

AI PLATFORM DEVELOPMENT

Nexora (Founder)

- Developing AI-driven operational analytics frameworks supporting process optimization, anomaly detection, forecasting, and decision-support systems
- Designing scalable data architectures for predictive analytics, reporting automation, and operational intelligence workflows
- Applying machine learning methodologies to improve workflow visibility, resource utilization, and data-driven operational performance